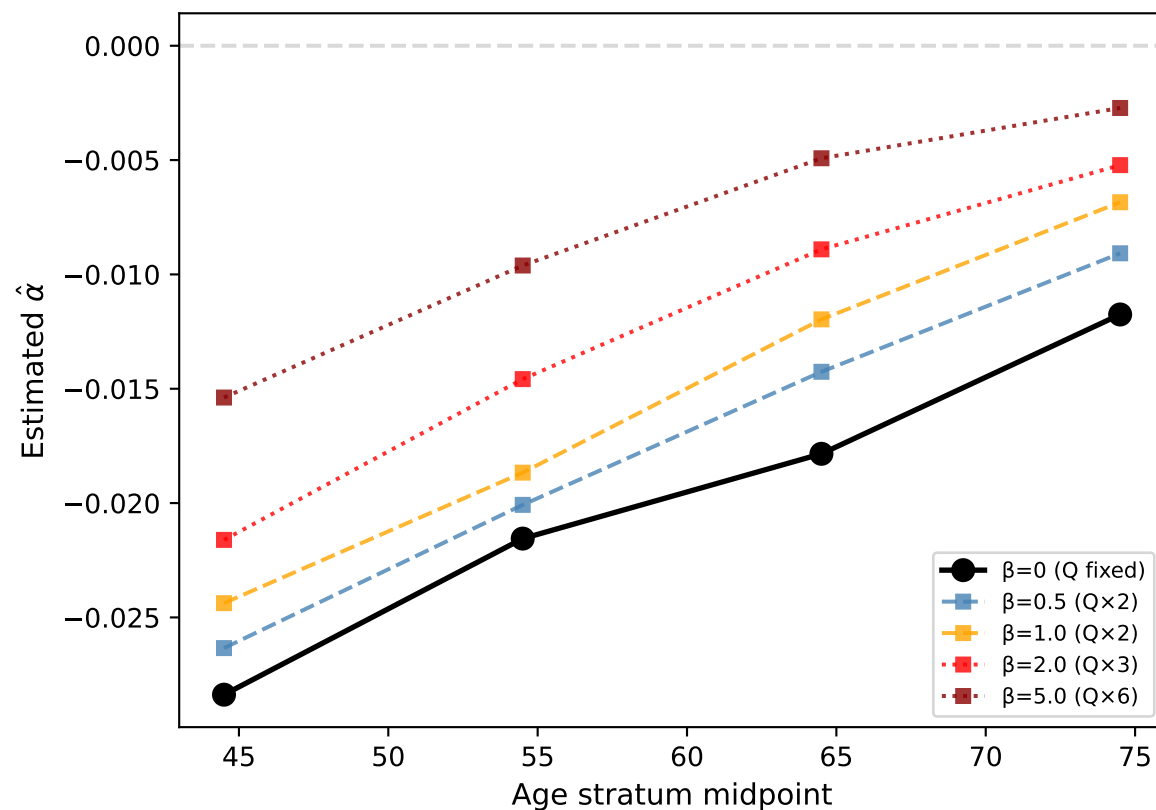
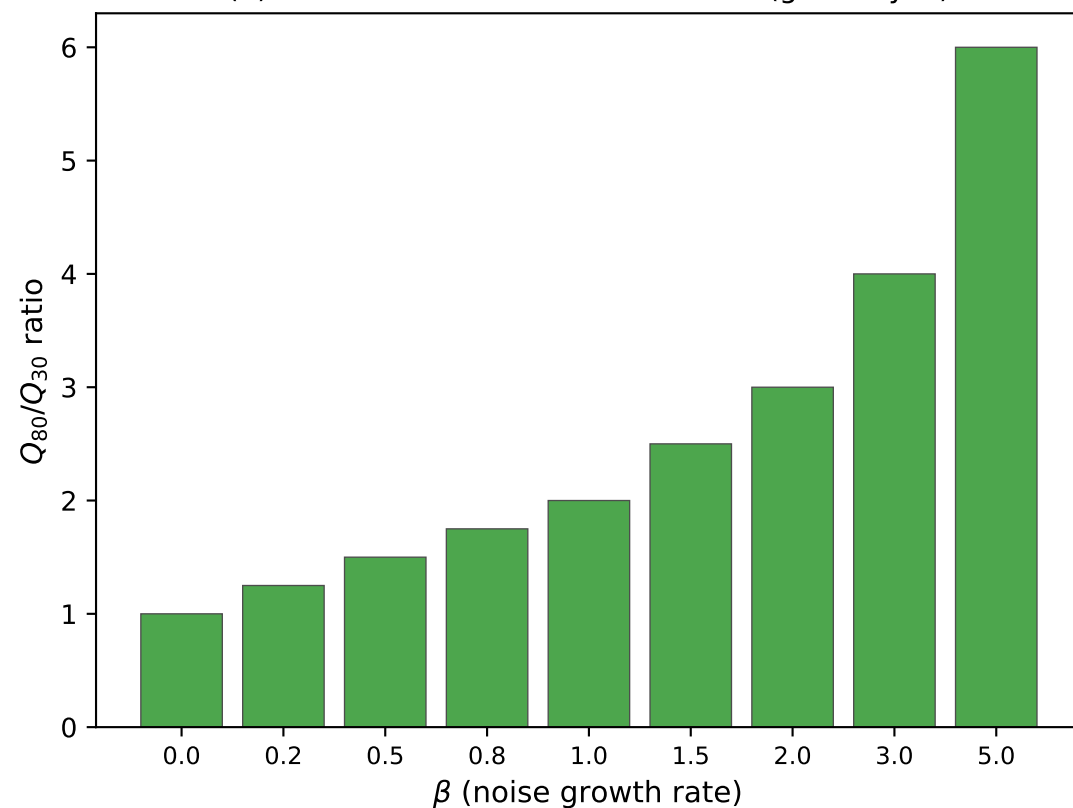


# Diffusion Covariance (Q) Sensitivity Analysis

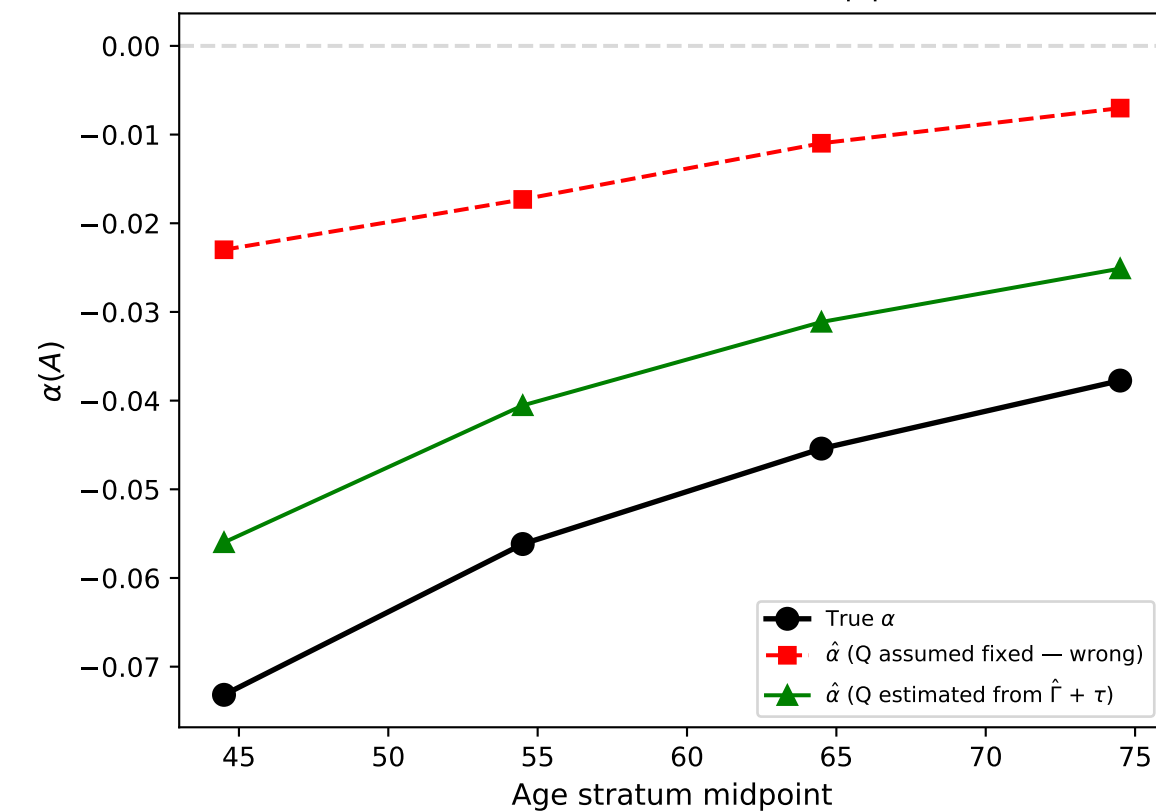
(a)  $\hat{\alpha}$  trend when Q assumed fixed but true Q varies



(b) Does monotone  $\hat{\alpha}$  trend survive? (green=yes)



(c) Q-estimation corrects the  $\hat{\alpha}$  pipeline



## Q-SENSITIVITY VERDICT

Threshold  $\beta^* \approx >5.0$   
(Q at age 80 =  $>6 \times$  Q at age 30)

If noise increases by less than  $>5 \times$  over the lifespan, the monotone  $\hat{\alpha}$  trend reflects genuine stability-margin erosion, not artifact of increasing noise.

### MITIGATION STRATEGIES:

- Estimate Q from  $\hat{\Gamma} + \text{literature } \tau$   
→ corrects bias even when Q varies
- Test-retest q from within-individual short-term variability (CGM, serial CRP)
- Report sensitivity across  $\beta = [0, 2]$  as standard robustness check

R4 NOTE:  $\lambda_{\max}(\hat{\Gamma})$  does not require Q specification — robust to Q misspecification by construction.